Counterfactual Reasoning for Responsibility Attribution in Probabilistic Multi-Agent Systems

Supplementary Material

1 Desirable Properties: Proofs

Lemma 1. *For a strategy profile $\vec{\sigma}$ and outcome φ, it holds:*

$$\Upsilon(\vec{\sigma}, \varphi) = v_{\vec{\sigma},\varphi}(\mathsf{Ag}) - v_{\vec{\sigma},\varphi}(\varnothing)$$

Proof. From the definition of attributable value and bCR degree, we have that:

$$v_{\vec{\sigma},\varphi}(\varnothing) = \min_{\vec{\sigma}_{-\varnothing}} \mathsf{Prob}(\{\rho \mid \rho \in \mathsf{Hist}_{\vec{\sigma}}(s^0)^{\leq k} \wedge \rho \models \varphi\}$$

and since $\mathsf{Ag} \setminus \varnothing = \mathsf{Ag}$, therefore $\vec{\sigma}_{\mathsf{Ag}} = \vec{\sigma}$, and we immediately obtain the lemma. □

Proposition 1 (Efficiency)**.** *For a strategy profile $\vec{\sigma}$ and outcome φ, it holds that: $\sum_{i\in\mathsf{Ag}} \mathbb{D}^i(\vec{\sigma}, \varphi) = \Upsilon(\vec{\sigma}, \varphi)$.*

Proof.

$$\sum_{i\in\mathsf{Ag}} \mathbb{D}^i(\vec{\sigma}, \varphi) = \sum_{i\in\mathsf{Ag}} \sum_{J\subseteq\mathsf{Ag}\setminus\{i\}} \frac{|J|!(|\mathsf{Ag}| - |J| - 1)!}{|\mathsf{Ag}|!} \cdot (v_{\vec{\sigma},\varphi}(J \cup \{i\}) - v_{\vec{\sigma},\varphi}(J))$$

$$= \sum_{J\subseteq\mathsf{Ag}} \sum_{i\in\mathsf{Ag}\setminus J} \frac{|J|!(|\mathsf{Ag}| - |J| - 1)!}{|\mathsf{Ag}|!} \cdot (v_{\vec{\sigma},\varphi}(J \cup \{i\}) - v_{\vec{\sigma},\varphi}(J))$$

Since for each subset J the term $\frac{|J|!(|\mathsf{Ag}|-|J|-1)!}{|\mathsf{Ag}|!}$ is the same for each player i not in J, and there are $|\mathsf{Ag}| - |J|$ such players, thus we can simplify the inner sum:

$$\sum_{i\in\mathsf{Ag}} \mathbb{D}^i(\vec{\sigma}, \varphi) = \sum_{J\subseteq\mathsf{Ag}} \frac{|J|!(|\mathsf{Ag}| - |J| - 1)!}{|\mathsf{Ag}|!} \sum_{i\in\mathsf{Ag}\setminus J} (v_{\vec{\sigma},\varphi}(J \cup \{i\}) - v_{\vec{\sigma},\varphi}(J))$$

Observe that the expression $\sum_{i\in\mathsf{Ag}\setminus J}(v_{\vec{\sigma},\varphi}(J \cup \{i\}) - v_{\vec{\sigma},\varphi}(J))$ is a *telescoping series/sum* with respect to all subsets J, thus we can simplify the inner sum further and obtain: $\sum_{i\in\mathsf{Ag}} \mathbb{D}^i(\vec{\sigma}, \varphi) = \sum_{J\subseteq\mathsf{Ag}} \frac{|J|!(|\mathsf{Ag}|-|J|-1)!}{|\mathsf{Ag}|!} \cdot (v_{\vec{\sigma},\varphi}(\mathsf{Ag}) - v_{\vec{\sigma},\varphi}(\varnothing))$. Since $\sum_{J\subseteq\mathsf{Ag}} \frac{|J|!(|\mathsf{Ag}|-|J|-1)!}{|\mathsf{Ag}|!} = 1$, we have:

$$\sum_{i\in\mathsf{Ag}} \mathbb{D}^i(\vec{\sigma}, \varphi) = v_{\vec{\sigma},\varphi}(\mathsf{Ag}) - v_{\vec{\sigma},\varphi}(\varnothing).$$

By Lemma 1, we obtain:

$$\sum_{i\in\mathsf{Ag}} \mathbb{D}^i(\vec{\sigma}, \varphi) = \Upsilon(\vec{\sigma}, \varphi).$$

□

Proposition 2 (Symmetry)**.** *For two agents $i, j \in \mathsf{Ag}$, if:*

$$\forall J \subseteq \mathsf{Ag} \setminus \{i, j\}, v_{\vec{\sigma},\varphi}(J \cup \{i\}) = v_{\vec{\sigma},\varphi}(J \cup \{j\})$$

then:

$$\mathbb{D}^i(\vec{\sigma}, \varphi) = \mathbb{D}^j(\vec{\sigma}, \varphi)$$

Proof. To prove this claim, we first introduce some notations. Denote by $\Pi(\mathsf{Ag})$ the set of all permutations of Ag. For every permutation $\pi \in \Pi(\mathsf{Ag})$, define

$$P_i(\pi) := \{j \in \mathsf{Ag} : \pi(j) < \pi(i)\},$$

that is the set of agents ahead of agent i when they are ordered according to permutation π. Using an alternative formulation for the Shapley value, we can express the bCR degree as follows.

$$\mathbb{D}^i(\vec{\sigma}, \varphi) = \frac{1}{|\mathsf{Ag}|!} \sum_{\pi\in\Pi(\mathsf{Ag})} (v_{\vec{\sigma},\varphi}(P_i(\pi) \cup \{i\}) - v_{\vec{\sigma},\varphi}(P_i(\pi))) \quad (1)$$

Define a function $f : \Pi(\mathsf{Ag}) \to \Pi(\mathsf{Ag})$ that maps each permutation over the set of agents to another permutation as follows. For every permutation π, the permutation $f(\pi)$ is identical to π except that it swaps agent i with agent j. Next we show that the following holds.

$$v_{\vec{\sigma},\varphi}(P_i(\pi) \cup \{i\}) - v_{\vec{\sigma},\varphi}(P_i(\pi)) = v_{\vec{\sigma},\varphi}(P_j(f(\pi)) \cup \{j\}) - v_{\vec{\sigma},\varphi}(P_j(f(\pi))) \quad (2)$$

There are two possible cases:

Case 1: agent i appears *before* agent j under permutation π, i.e., $j \notin P_i(\pi)$. Then $P_i(\pi) = P_j(f(\pi))$, as such, $v_{\vec{\sigma},\varphi}(P_i(\pi)) = v_{\vec{\sigma},\varphi}(P_j(f(\pi)))$. Morever, since both $P_i(\pi)$ and $P_j(f(\pi))$ do not contain i nor j, it holds that:

$$v_{\vec{\sigma},\varphi}(P_i(\pi) \cup \{i\}) = v_{\vec{\sigma},\varphi}(P_j(f(\pi)) \cup \{j\})$$

Therefore, these two equalities imply Equation (2).

Case 2: agent i appears *after* agent j under permutation π, i.e., $j \in P_i(\pi)$. Then $P_i(\pi) \cup \{i\} = P_j(f(\pi)) \cup \{j\}$, as

such, $v_{\vec{\sigma},\varphi}(P_i(\pi)\cup\{i\}) = v_{\vec{\sigma},\varphi}(P_j(f(\pi))\cup\{j\})$. Moreover, $P_i(\pi)\backslash\{j\} = P_j(f(\pi))\backslash\{i\}$, and since i and j are symmetric, it holds that:

$$v_{\vec{\sigma},\varphi}((P_i(\pi)\setminus\{j\})\cup\{j\}) = v_{\vec{\sigma},\varphi}((P_j(f(\pi))\setminus\{i\})\cup\{i\}),$$

i.e., $v_{\vec{\sigma},\varphi}(P_i(\pi)) = v_{\vec{\sigma},\varphi}(P_j(f(\pi)))$. Therefore, these two equalities also imply Equation (2).

From Equation (1) and Equation (2), we have

$$\mathbb{D}^i(\vec{\sigma},\varphi) = \frac{1}{|\mathsf{Ag}|!}\sum_{\pi\in\Pi(\mathsf{Ag})} (v_{\vec{\sigma},\varphi}(P_j(f(\pi))\cup\{j\}) - v_{\vec{\sigma},\varphi}(P_j(f(\pi)))) \quad (3)$$

Since f is bijective, thus $\Pi(\mathsf{Ag}) = \{f(\pi) : \pi \in \Pi(\mathsf{Ag})\}$, and by setting $\mu = f(\pi)$ we obtain

$$\mathbb{D}^i(\vec{\sigma},\varphi) = \frac{1}{|\mathsf{Ag}|!}\sum_{\mu\in\Pi(\mathsf{Ag})} (v_{\vec{\sigma},\varphi}(P_j(\mu)\cup\{j\}) - v_{\vec{\sigma},\varphi}(P_j(\mu))) \quad (4)$$

And, consequently $\mathbb{D}^i(\vec{\sigma},\varphi) = \mathbb{D}^j(\vec{\sigma},\varphi)$. □

Proposition 3 (Null Player)**.** *If for all $J \subseteq \mathsf{Ag}\backslash\{i\} : v_{\vec{\sigma},\varphi}(J\cup\{i\}) = v_{\vec{\sigma},\varphi}(J)$, then $\mathbb{D}^i(\vec{\sigma},\varphi) = 0$.*

Proof. If for a given $J \subseteq \mathsf{Ag}\backslash\{i\} : v_{\vec{\sigma},\varphi}(J\cup\{i\}) = v_{\vec{\sigma},\varphi}(J)$, then we have $(v_{\vec{\sigma},\varphi}(J\cup\{i\}) - v_{\vec{\sigma},\varphi}(J)) = 0$. Since it holds for every $J \in \mathsf{Ag}\setminus\{i\}$, then

$$\mathbb{D}^i(\vec{\sigma},\varphi) = \sum_{J\subseteq\mathsf{Ag}\backslash\{i\}} \frac{|J|!(|\mathsf{Ag}|-|J|-1)!}{|\mathsf{Ag}|!}\cdot 0 = 0.$$

□

Proposition 4 (Sub-additivity)**.** *Given $\vec{\sigma}$, for any two outcome φ_1 and φ_2, and any $i \in \mathsf{Ag}$:*

$$\mathbb{D}^i(\vec{\sigma},\varphi_1\vee\varphi_2) \le \mathbb{D}^i(\vec{\sigma},\varphi_1) + \mathbb{D}^i(\vec{\sigma},\varphi_2)$$

Proof. For any fixed coalition $A \subseteq \mathsf{Ag}$ and any strategy profile $(\vec{\sigma}_A,\vec{\sigma}_{-A})$, we have:

$$\begin{aligned}&\mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{(\vec{\sigma}_A,\vec{\sigma}_{-A})}(s^0)^{\le k}\wedge\rho\models(\varphi_1\vee\varphi_2)\})\\ &\le \big(\mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{(\vec{\sigma}_A,\vec{\sigma}_{-A})}(s^0)^{\le k}\wedge\rho\models\varphi_1\})\\ &+ \mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{(\vec{\sigma}_A,\vec{\sigma}_{-A})}(s^0)^{\le k}\wedge\rho\models\varphi_2\})\big) \quad (5)\end{aligned}$$

Since the value of the sum at the joint minimiser is no more than the sum of the individual minima, which may occur at different points:

$$\min_x[f(x)+g(x)] \le \min_x f(x) + \min_x g(x)$$

taking the minimum over $\vec{\sigma}_{-A}$ on both sides of Equation (5), we have:

$$v_{\vec{\sigma},\varphi_1\vee\varphi_2}(A) \le v_{\vec{\sigma},\varphi_1}(A) + v_{\vec{\sigma},\varphi_2}(A) \quad \text{for all } A\subseteq\mathsf{Ag}$$

By monotonicity of the Shapley value under pointwise function, we obtain the result. □

Proposition 5 (Exact Additivity)**.** *Let φ_1 and φ_2 be two outcome formulas such that they are disjoint and avoidable, then for any $i \in \mathsf{Ag}$:*

$$\mathbb{D}^i(\vec{\sigma},\varphi_1\vee\varphi_2) = \mathbb{D}^i(\vec{\sigma},\varphi_1) + \mathbb{D}^i(\vec{\sigma},\varphi_2)$$

Proof. By the disjointness assumption on φ_1 and φ_2, and the standard property of probability over disjoint events, we have:

$$\begin{aligned}&\mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{\vec{\sigma}_A,\vec{\sigma}_{-A}}(s^0)^{\le k}\wedge\rho\models(\varphi_1\vee\varphi_2)\})\\ &= \mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{\vec{\sigma}_A,\vec{\sigma}_{-A}}(s^0)^{\le k}\wedge\rho\models\varphi_1\})\\ &+ \mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{\vec{\sigma}_A,\vec{\sigma}_{-A}}(s^0)^{\le k}\wedge\rho\models\varphi_2\}) \quad (6)\end{aligned}$$

Applying the pessimistic valuation to both sides of Equation (6), we obtain:

$$v_{\vec{\sigma},\varphi_1\vee\varphi_2}(A) \ge v_{\vec{\sigma},\varphi_1}(A) + v_{\vec{\sigma},\varphi_2}(A) \quad \text{for all } A\subseteq\mathsf{Ag}$$

Since $\varphi_1\vee\varphi_2$ is avoidable (thus $\mathbb{D}^i(\vec{\sigma},\varphi_1\vee\varphi_2) > 0$) by the assumption, by monotonicity of the Shapley value under pointwise function, we have:

$$\mathbb{D}^i(\vec{\sigma},\varphi_1\vee\varphi_2) \ge \mathbb{D}^i(\vec{\sigma},\varphi_1) + \mathbb{D}^i(\vec{\sigma},\varphi_2)$$

By combining with Proposition 4, we obtain the proposition. □

Proposition 6 (Monotonicity)**.** *If $\varphi_1 \Rightarrow \varphi_2$ and φ_2 is avoidable, then for any $i \in \mathsf{Ag}$ and a given $\vec{\sigma}$:*

$$\mathbb{D}^i(\vec{\sigma},\varphi_1) \le \mathbb{D}^i(\vec{\sigma},\varphi_2)$$

Proof. For every coalition $A \subseteq \mathsf{Ag}$, and any strategy profile $(\vec{\sigma}_A,\vec{\sigma}_{-A})$, the implication $\varphi_1 \Rightarrow \varphi_2$ ensures:

$$\begin{aligned}&\mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{\vec{\sigma}_A,\vec{\sigma}_{-A}}(s^0)^{\le k}\wedge\rho\models\varphi_1\})\\ &\le \mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{\vec{\sigma}_A,\vec{\sigma}_{-A}}(s^0)^{\le k}\wedge\rho\models\varphi_2\})\end{aligned}$$

Therefore, taking the minimum over $\vec{\sigma}_{-A}$, we get:

$$v_{\vec{\sigma},\varphi_1}(A) \le v_{\vec{\sigma},\varphi_2}(A) \quad \text{for all } A\subseteq\mathsf{Ag}$$

Since φ_2 is avoidable (thus $\mathbb{D}^i(\vec{\sigma},\varphi_2) > 0$) by the assumption, and the Shapley value is monotonic under pointwise comparison of value functions, it follows that: $\mathbb{D}^i(\vec{\sigma},\varphi_1) \le \mathbb{D}^i(\vec{\sigma},\varphi_2)$. □

2 Computing Responsibility Degree and Model Checking PATL-SR

Lemma 2. *Given a CSG $\mathcal{G}$, a coalition A, strategy profile $\vec{\sigma}$, and a path formula ψ, computing $\mathbb{D}^i(\vec{\sigma}_A,\psi)$ can be done in polynomial space.*

Proof. To show that it can be computed in polynomial space, we propose Algorithm 1. We show that the algorithm runs in PSPACE.

We first examine lines 5 and 6. Observe that computing:

$$\min_{\vec{\sigma}_{-J}} \mathsf{Prob}(\{\rho \mid \rho\in\mathsf{Hist}_{(\vec{\sigma}_J,\vec{\sigma}_{-J})}(s^0)^{\le k}\wedge\rho\models_{\mathcal{M}}\psi\})$$

corresponds to model checking a *probabilistic* CTL (PCTL) formula over a *Markov Decision Process* (MDP) [Baier and

Algorithm 1 degree-BCR: Calculate $\mathbb{D}^i(\vec{\sigma}_A, \psi)$

1: **Input:** $\mathcal{M}, i, \vec{\sigma}, A, \psi, k$
2: **Output:** $\mathbb{D}^i(\vec{\sigma}_A, \psi)$
3: $d \leftarrow 0$
4: **for** each $J \subseteq A \setminus \{i\}$ **do**
5: $\quad v_1 \leftarrow \min_{\vec{\sigma}_{-(J\cup\{i\})}} \mathsf{Prob}(\{\rho \mid \rho \in \mathsf{Hist}_{(\vec{\sigma}_{(J\cup\{i\})}, \vec{\sigma}_{-(J\cup\{i\})})}(s^0)^{\leq k} \wedge \rho \models \psi\})$
6: $\quad v_2 \leftarrow \min_{\vec{\sigma}_{-J}} \mathsf{Prob}(\{\rho \mid \rho \in \mathsf{Hist}_{(\vec{\sigma}_J, \vec{\sigma}_{-J})}(s^0)^{\leq k} \wedge \rho \models \psi\})$
7: $\quad d \leftarrow d + \frac{|J|!(|\mathsf{Ag}|-|J|-1)}{|\mathsf{Ag}|!} \cdot (v_1 - v_2)$
8: **end for**
9: **Return** d

Katoen, 2008]. When we apply the (partial) strategy profile $\vec{\sigma}_J$, we obtain an MDP $\mathcal{G}_{\vec{\sigma}_J}$.

We then compute the quantitative value of the PCTL formula $\mathbf{P}_{\min=?}\psi$, which represents the minimum probability value of ψ in the MDP $\mathcal{G}_{\vec{\sigma}_J}$. This can be solved in polynomial time using linear programming and value iteration method [Baier and Katoen, 2008]. Therefore, lines 5 and 6 are solvable in polynomial time.

Furthermore, the values $|J|!$ and $|\mathsf{Ag}|!$ require $O(n \log n)$ bits. Since for each subset J, we only need to keep the current intermediate result in memory, and each intermediate result is of polynomial size, the PSPACE membership follows immediately. □

Theorem 1. *Model checking* PATL-SR *is in PSPACE.*

Proof. The proof consists of two parts.

First, we show that model checking PATL-SR formula without $\mathbf{D}$ operators is in PSPACE. This follows from the fact that the fragment of PATL-SR without $\mathbf{D}$ operators corresponds to the logic rPATL proposed in [Kwiatkowska *et al.*, 2019], whose model checking over CSGs is shown to be in PSPACE.

Second, by Lemma 2, model checking PATL-SR formulae of the form $\langle A\rangle\mathbf{D}_{\bowtie d}[\mathsf{bCR}_i(\vec{\sigma}, \psi)]$ can also be done in PSPACE. Therefore, the PATL-SR model checking problem can be solved in PSPACE. □

3 CR-aware Strategic Decision-Making: Proofs

Proposition 7 (Quasi-Concavity of Utility)**.** *The utility function u_i is quasi-concave if the following conditions hold:*

1) *linearity of reward: the expected reward is linear in $\vec{\sigma}_i$;*
2) *linearity of responsibility contribution: the responsibility-valuation function $v_{\vec{\sigma},\varphi}(A)$ is linear in $\vec{\sigma}_i$ for all coalition $A \subseteq \mathsf{Ag}$;*
3) *fixed strategies of others: the strategy profile $\vec{\sigma}_{-i}$ of other agents is fixed.*

Proof. Under condition 1) and 2), both the expected reward and the BCR degree term are linear functions of $\vec{\sigma}_i$. A linear function is trivially quasi-concave (and also quasi-convex). Since $\vec{\sigma}_{-i}$ is fixed, we are examining a function $u_i(\vec{\sigma}_i)$ over a convex domain (probability simplex), which is a linear combination of linear terms. Hence u_i is quasi-concave in $\vec{\sigma}_i$ under these conditions. □

Proposition 8. *In a game where the formula φ is a reachability or safety formula, the quasi-concavity property (Proposition 7) holds.*

Proof. For the responsibility component, note that when φ is either *reachability* or *safety*, pure memoryless strategies are sufficient to satisfy it [Chatterjee and Henzinger, 2012]. Thus, the probability of satisfying φ can be expressed as a linear polynomial over strategy parameters x_i. Consequently, $v_{\vec{\sigma},\varphi}(A)$ is linear in $\vec{\sigma}_i$, and thus $v_{\vec{\sigma},\varphi}(A \cup \{i\}) - v_{\vec{\sigma},\varphi}(A)$ is also linear in $\vec{\sigma}_i$. Therefore, $\mathbb{D}^i(\vec{\sigma}, \varphi)$ is linear.

For the reward component, since pure memoryless strategies are sufficient, the reward can be represented by a linear function. This linearity condition also trivially extends to strategy profiles. □

References

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